

DYNAMIC STRETCHES

Dynamic stretching can prepare muscles and joints in a more specific manner than static stretching because the body is performing the motions it will likely repeat in the poststretch activity. During dynamic stretching, you're active the entire time you are preparing for activity. This constant activity raises and maintains a higher body temperature, and, thus, becomes an important part of your warm-up. In contrast, a person usually experiences a slight drop or no change in body temperature when doing static stretches. Because dynamic stretches more precisely mimic body movements, they prepare the muscles for the actual movements. This enhances the training effect placed on the nerves and muscles. Dynamic stretching is less likely to cause the performance decrements that can arise from the other stretching methods—ballistic, proprioceptive neuromuscular facilitation, and static—described in chapter 1.

As you do for any other activity, you must follow specific guidelines and principles when performing dynamic stretches:

- An effective warm-up that includes dynamic stretches should last 10 to 15 minutes or 10 to 20 repetitions per stretch.
- Observe your initial body position when you perform a particular activity, and then make sure you start the dynamic stretch from the same initial position.
- Discern the range of motion that each joint travels. The dynamic stretch range of motion should only slightly exceed the range of motion of the activity for which you are preparing.
- The dynamic stretch should closely replicate the movements used during the activity. Use proper technique and use all of the muscles normally used during the activity. If the dynamic stretches mimic a specific sport skill, such as a high knee lift, the stretches should use the specific factors of the skill. If you are careful to

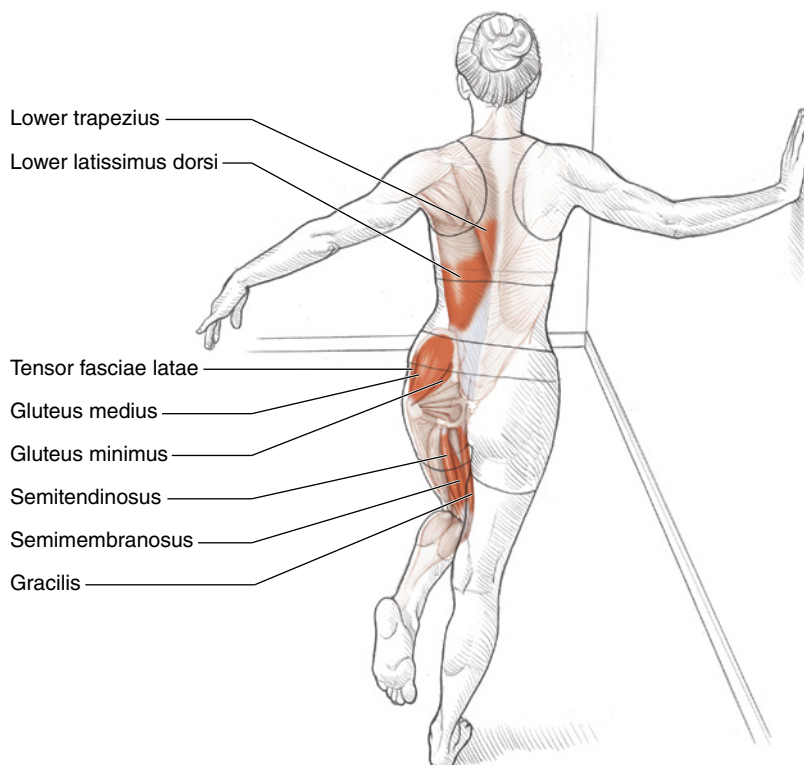
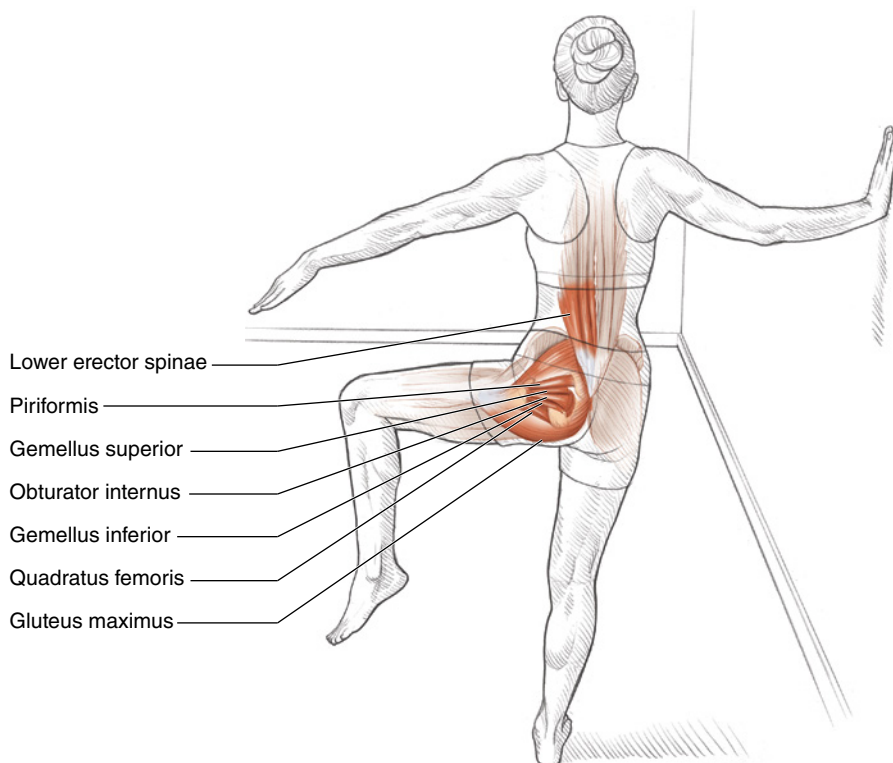
mimic the skill as closely as possible, you enhance the learning of the skill specificities and diminish the chances of introducing improper techniques.

- Performing the movements too fast or by generating enough momentum that you greatly exceed the joint range of motion enhances the risk of injury.
- When doing dynamic stretches, you can either perform repetitions in the same place or travel for a set distance. Whether you remain in place or move, you should start each stretch slowly and progressively increase the range of motion and movement speed with each repetition. For instance, if you are moving over a distance, start with a walk, proceed to a skip, and finally end with a run.
- Dynamic stretches can be done singularly or in combination. Combining two or more stretches provides variety in your program and enables you to better mimic more complex skills.

In summary, each dynamic stretch should include 10 to 20 repetitions done either in place or traveling over a set distance; you should progressively increase the range of motion and the speed of movement; the muscles should be contracted throughout the stretch; you should use proper technique for each repetition, just as you would normally perform the action; and you must ensure the movements are completely controlled by performing deliberate actions with no bouncing.

People preparing for competitive or recreational activities can use the following dynamic stretches as a preexercise warm-up. In most cases, they are helpful for almost any sport. These dynamic stretches concentrate on the major muscle groups in the body and are very easy to execute. You will find more enjoyment in your training or activity if you include these preexercise dynamic stretches in your program. In chapter 11, you will find specific programs and recommendations for a variety of sporting events. You have multiple options from which to choose when deciding which stretching exercises best fit your purposes.

DYNAMIC HIP EXTERNAL AND INTERNAL ROTATOR STRETCH



Execution

1. Stand upright on the right leg, with the knee straight. Stand with the right side facing a supporting surface such as a wall, the edge of a corner, or a doorway. Hold on to the supporting object at shoulder height.
2. Bend the left knee and hip slightly, and let the left leg hang down in a relaxed manner as the starting point of this dynamic stretch.
3. Swing and rotate your left bent leg around your hip in a circular motion, inside and outside directions, in a dynamic manner.
4. Keep the trunk upright, and allow the circular motion to take place around the hip joint.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles in external rotation: Left gluteus maximus, left gluteus medius, left gluteus minimus, left piriformis, left gemellus superior, left gemellus inferior, left obturator externus, left obturator internus, left quadratus femoris, lower left erector spinae (iliocostalis, longissimus, spinalis)

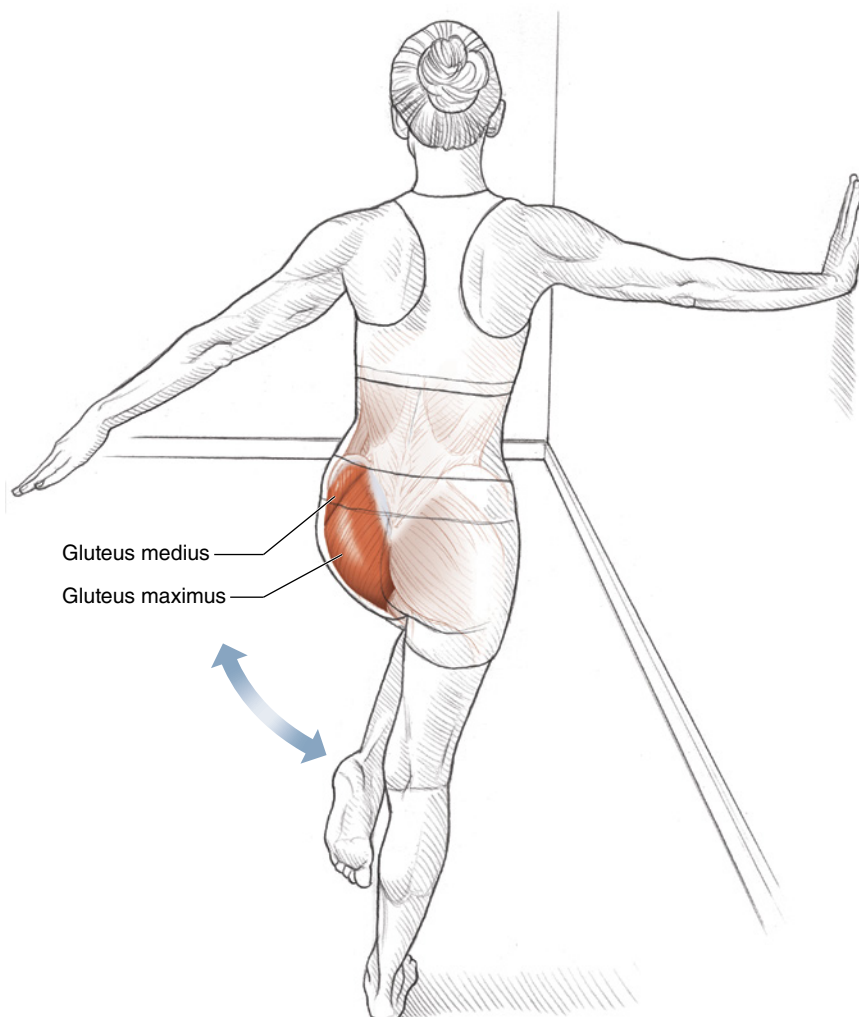
Most-stretched muscles in internal rotation: Left gluteus medius, left gluteus minimus, left tensor fasciae latae, left semitendinosus, left semimembranosus, left gracilis, lower left latissimus dorsi, lower left trapezius

Stretch Notes

The hip external rotator muscles are located in the deep tissue of the hip just under the gluteus maximus muscle. These particular muscles can become sore or tight when unusual stress is placed on them or after engaging in activities that are not common in daily routines. Soreness or tightness is often caused by extensive use of the hip external and internal rotator muscles in activities such as ice skating, in-line skating, or the skating style of cross-country skiing. Many other activities, such as an impromptu game of soccer requiring sprinting, jumping, and making sudden changes of direction, can easily result in uncomfortable or painful muscles later on.

On subsequent days, if soreness or tightness is still present in these particular muscles, before starting activities that require hip external or internal rotational movements, use this dynamic stretch to warm up. This dynamic stretch increases the effectiveness of muscular movements and enhances total performance in many sporting activities.

DYNAMIC HIP ADDUCTOR AND ABDUCTOR STRETCH



Execution

1. Stand upright on the right leg, with the knee straight. Stand with the right side facing a supporting surface such as a wall, the edge of a corner, or a doorway. Hold on to the supporting object at shoulder height.
2. Bend the left knee and hip slightly, and let the left leg hang down in a relaxed manner as the starting point of this dynamic stretch.
3. Swing your left leg in front of you from side to side in a dynamic manner, with just enough clearance to avoid hitting the right leg. Make sure the knee of the swinging leg stays slightly bent.

4. Keep the trunk upright, and allow the movement to take place in the hip joint by using the adductor muscles located inside the thigh and hip and the abductor muscles located outside the thigh and hip.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles on inside of thigh: Left gracilis, left adductor magnus, left adductor longus, left adductor brevis, left pectineus, middle and lower left sartorius, left semitendinosus, left semimembranosus

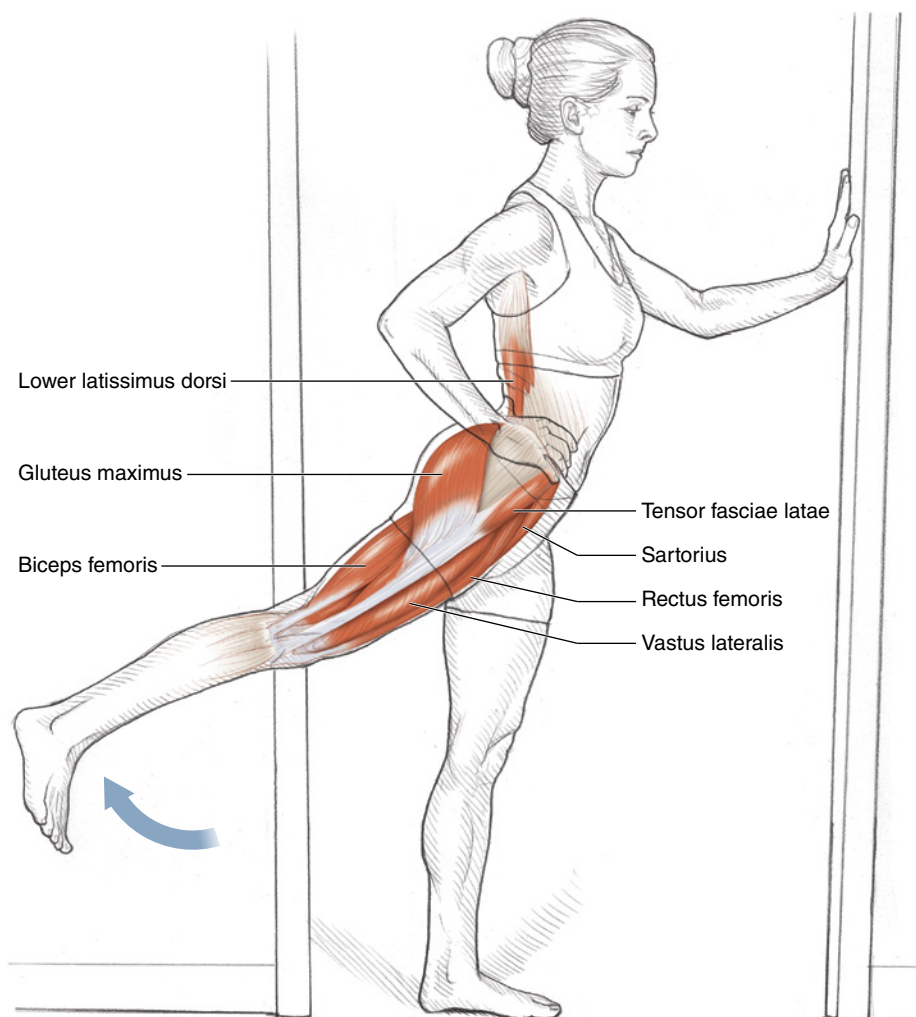
Most-stretched muscles on outside of thigh: Left gluteus medius, left gluteus minimus, left gluteus maximus, left tensor fasciae latae, upper left sartorius

Stretch Notes

The muscles on the medial (inner) side and lateral (outer) side of the hip and thigh are fairly large. As a group, they are called the adductor and abductor muscles, respectively. These muscles are responsible for hip adduction (bringing the leg toward the midline of the body) and abduction (moving the leg away from the midline of the body). They also keep the legs centered under the body and are used as stabilizer muscles in the performance of daily activities. Certain unusual movements or activities, such as repeated stair climbing or hiking uphill or downhill, can cause the muscles in this region to feel sore or fatigued, a condition that could easily continue on subsequent days. Regular stretching most likely will alleviate some of the symptoms. It is strongly recommended that you stretch the adductor and abductor muscles both before and after participating in sports or other strenuous activities to help prevent injuries or symptoms.

This is a helpful and effective preexercise dynamic stretch for people who feel muscular pain or general stiffness in the inner or outer thigh. Pain in any region of the body is often a result of muscular soreness. When muscles are sore, they often feel stiff as well. A person with this condition has the tendency to limit the range of motion of the affected muscles in order to avoid pain. Therefore, normal daily activities can be significantly affected depending on the severity of the pain. Rather than avoiding movement, a person experiencing muscular soreness or tightness should specifically try to move and stretch the injured muscles in a dynamic manner before starting an exercise routine. Performing this dynamic stretch for the hip adductors and abductors will increase flexibility and warmth in these muscle groups just before the activity, which in turn will lessen the likelihood or severity of injury and also possibly increase exercise capacity.

DYNAMIC HIP FLEXOR AND EXTENSOR STRETCH



Execution

1. Stand upright facing one side of a doorway. Stand on the left leg, with the knee straight. Hold on to the supporting object at the height of your shoulder.
2. Bend the right knee and hip slightly, and let the right leg hang down in a relaxed manner as the starting point of this dynamic stretch.
3. Keeping the leg slightly bent, swing your right leg straight forward and backward in a dynamic manner so that it swings parallel to the opening of the doorway.
4. Keep the trunk upright, and allow the movement to take place in front of and behind the hip joint, using the flexor and extensor muscles of the hip.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles on front of hip: Right quadriceps (rectus femoris, vastus lateralis, vastus intermedius, vastus medialis), right tensor fasciae latae, right sartorius

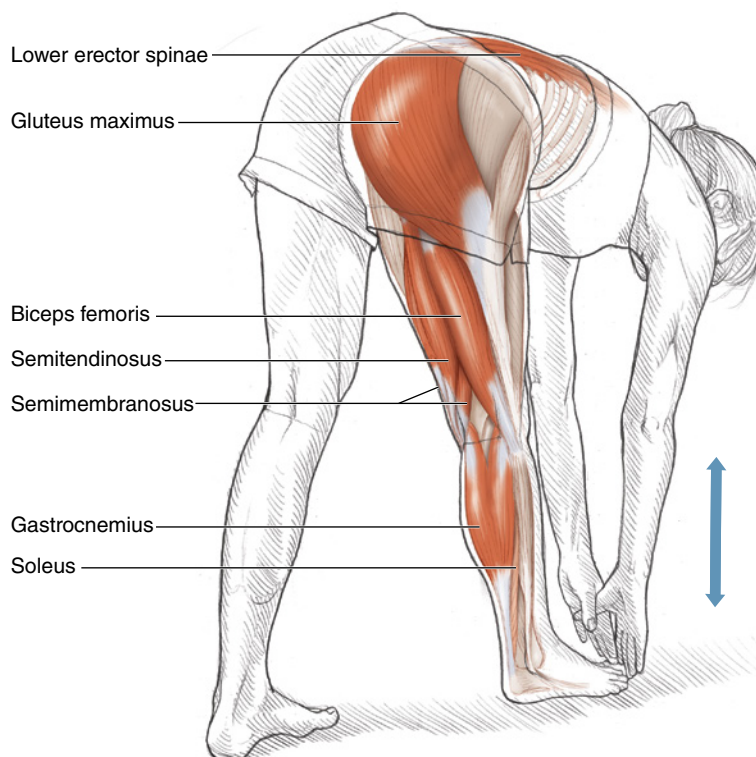
Most-stretched muscles on back of hip: Right gluteus maximus, right hamstring (semitendinosus, semimembranosus, biceps femoris), lower right erector spinae (iliocostalis, longissimus, spinalis), lower right latissimus dorsi

Stretch Notes

The hip flexor and extensor muscles are used extensively in most sports. These muscles often fatigue first, and as a consequence, performance decreases. Muscle soreness and tightness follow as the athlete continues to use these muscles. If they are not stretched properly, most likely the hamstrings and quadriceps will tighten up even more. Tight hamstrings and quadriceps are common among exercisers who significantly increase speed, the distance run, or the amount of uphill climbing during training. Tightness in the muscles can ease during exercise as the muscles get warmer, but when the athlete stops, the pain can return. Thus, it is especially important to stretch properly after exercise.

It is equally important to do dynamic preexercise stretches before engaging in your regular exercise routine. This dynamic stretch for the hip flexors and extensors will alleviate some of the problems you might encounter as you exercise these muscles extensively. We recommend performing this stretch as a warm-up before doing higher-intensity workouts.

DYNAMIC STANDING KNEE FLEXOR STRETCH



Execution

1. Stand upright with the right heel 1 to 2 feet (30 to 60 cm) ahead of the left toes.
2. Keeping the right knee straight and the left knee slightly bent, bend the trunk over toward the right knee.
3. Reach the hands toward the right foot.
4. Do the stretching in a dynamic manner by bobbing.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right hamstring (semitendinosus, semimembranosus, biceps femoris), right gluteus maximus, right gastrocnemius, lower right erector spinae (iliocostalis, longissimus, spinalis)

Less-stretched muscles: Right soleus, right plantaris, right popliteus, right flexor digitorum longus, right flexor hallucis longus, right tibialis posterior

Stretch Notes

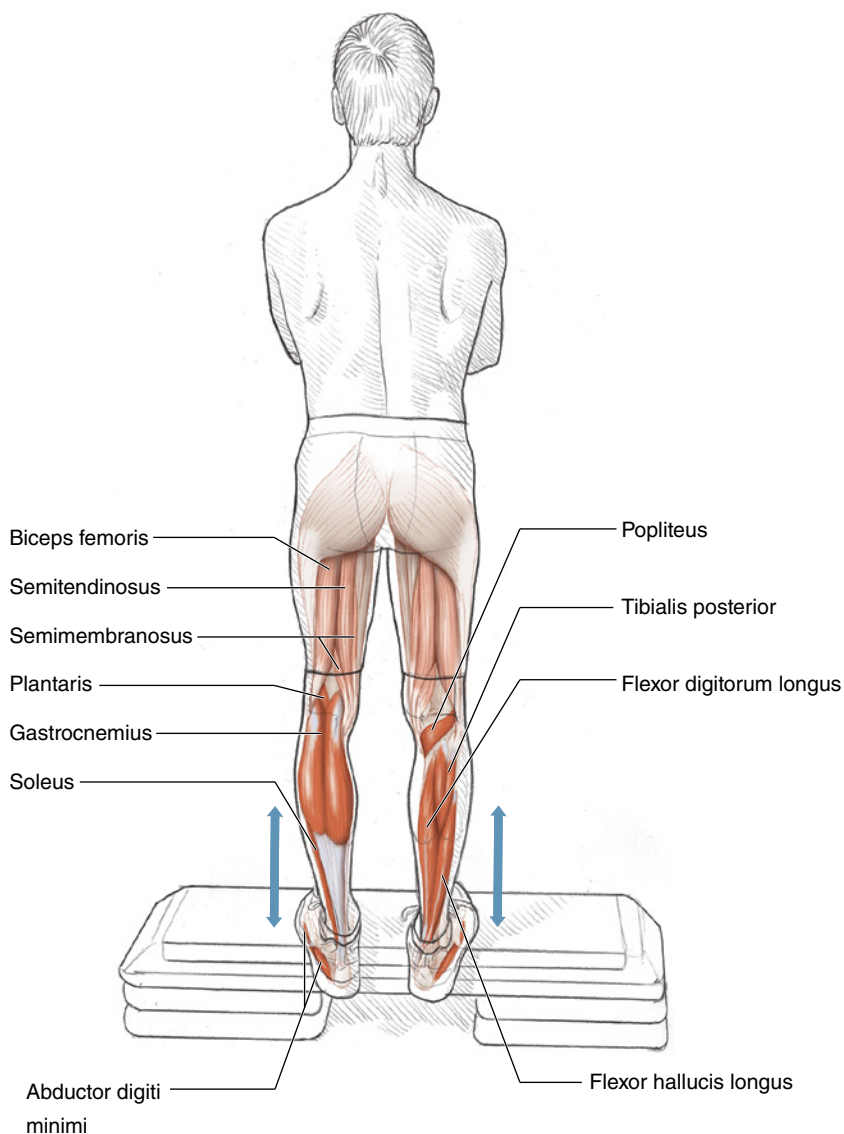
When you start participating in a sport and do not stretch properly, it is more likely that your hamstrings will tighten up. Tight hamstrings are common among many athletes and those who participate in recreational activities. Tightness in these muscles can ease during exercise as the muscles get warmer, but when the athlete stops, the pain can return.

Tightness is often an indicator of minor or major muscle strains, a common occurrence mainly felt postexercise. In addition, muscle strength imbalances, in which the knee extensors are stronger or the gluteal muscles are weaker than the hamstrings, will also cause tightness. Thus, it is especially important to stretch properly after exercise because this is when the muscles are warm and more receptive to stretching.

This is the most common preexercise stretch for the hamstring and calf muscles. The hamstrings are used in most activities, and you might feel discomfort in these muscles from your previous exercise session. In any type of fitness activity, minor aches and tightness in the hamstrings are possible. The optimal time to lightly stretch these muscles is just before you start another exercise session. In most cases, light dynamic stretches will relieve those uncomfortable symptoms, and you will feel so much better after engaging in these dynamic stretches.

For the best results, try to keep the right knee straight, and bend the torso directly from the hip. It is also important to keep the back as straight as possible. If you have tightness on the outer side of the hamstring muscles, turn the right foot slightly out and bend the head and trunk more toward the medial (inner) side of the right knee to increase the stretch of the biceps femoris. On the other hand, turning the right foot slightly in and bending the head and trunk more toward the lateral (outer) side of the knee will increase the stretch of the semitendinosus and semimembranosus muscles, located on the inner side of the hamstring muscles.

DYNAMIC PLANTAR FLEXOR STRETCH



Execution

1. Stand upright on the edge of a stair or beam, with the midsection of both feet on the edge.
2. Hold on to a support with at least one hand, and keep the knees straight.
3. Lower the heels as far as possible, and execute this stretch in a dynamic, bobbing up and down manner. Make sure to return the foot to the neutral position on each upward movement, and lower the heels as low as comfortably possible on the downward movements.

Muscles Stretched

Most-stretched muscles: Gastrocnemius, soleus, plantaris, popliteus, flexor digitorum longus, flexor digitorum brevis, flexor hallucis longus, flexor hallucis brevis, tibialis posterior, quadratus plantae, flexor digiti minimi brevis, abductor digiti minimi, abductor hallucis

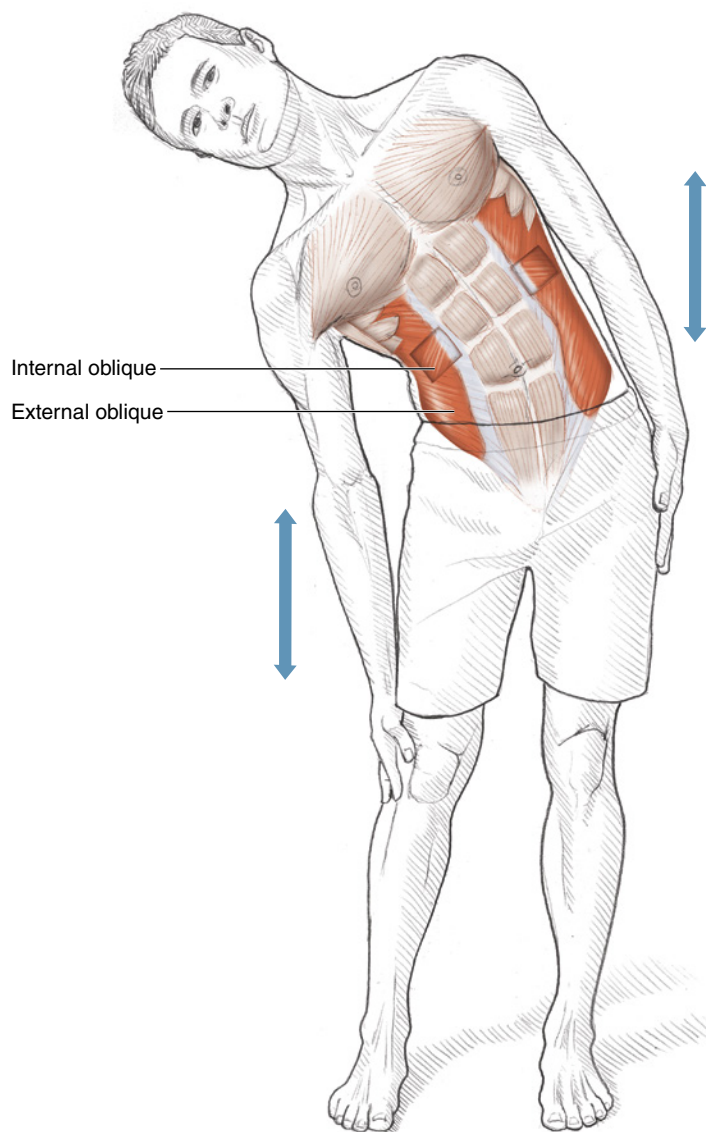
Less-stretched muscles: Hamstrings (semitendinosus, semimembranosus, biceps femoris)

Stretch Notes

This stretch is often performed after exercise, but it is also highly recommended as a preexercise stretch. The calf muscles are in heavy use most of the day. They take most of the load during walking, running, and jumping activities. Naturally, they become overworked, which sometimes leads to serious problems such as tendinitis or even muscle tears. As a preexercise stretch, this dynamic stretch for the plantar flexors will alleviate some of the problems you might encounter as you exercise these muscles extensively. We recommend performing this stretch as a warm-up before doing higher-intensity workouts. You should also add a postexercise static plantar flexor stretch to your overall training program.

It is more comfortable to do this stretch while wearing shoes. Always support the body. Not supporting the body could cause the muscles to contract and not stretch. Do not overstretch these muscles when doing this exercise. Start easy, and slowly progress to a higher intensity level.

DYNAMIC TRUNK LATERAL FLEXION STRETCH



Execution

1. Stand upright with your feet shoulder-width apart.
2. Let the arms hang down by your sides.
3. With the help of your arms, bend your torso laterally back and forth in a dynamic manner. As you bend to the right, slid the right arm down the right thigh toward your knee. Return to the starting position, and then bend to the left. Alternate between right and left.
4. Allow all dynamic movement to take place on the lateral side of the trunk.

Muscles Stretched

Most-stretched muscles: External oblique, internal oblique, intertransversarii, multifidus, quadratus lumborum, rotatores

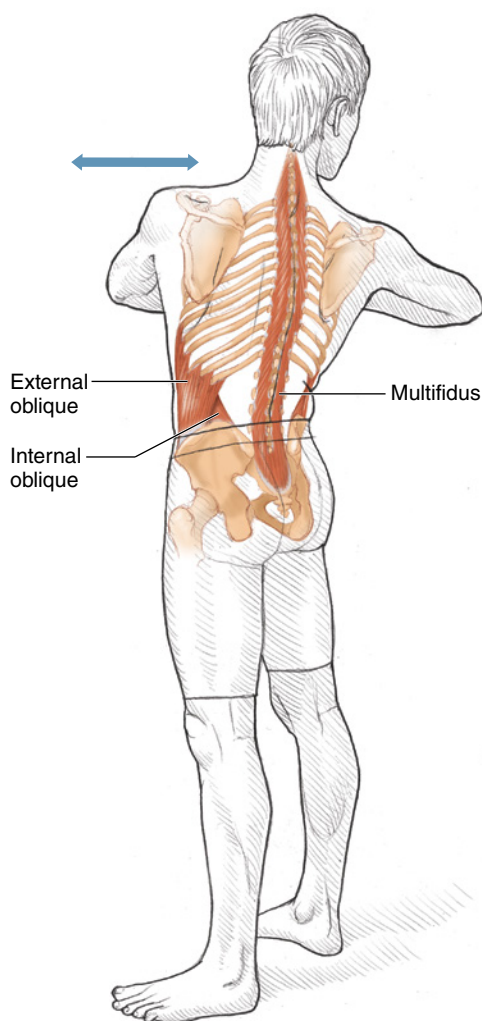
Stretch Notes

Trunk lateral flexion stretching movements are often used in regular routines for nonspecific sports activity. You bend your trunk regularly in different directions many times a day. You might feel unusual tightness or soreness in these muscles and simply want relief from these discomforts. Twisting the trunk goes along with lateral trunk flexion. These two muscle movements involve the muscles of the trunk extensors, flexors, and lateral flexors. Improved range of motion of all lower-trunk muscles can increase the range of motion in trunk lateral flexion and improve performance in activities that involve nonspecific sports actions.

These core muscle groups are also often used as stabilizer muscles that allow other muscles to apply force. Thus, it is important to keep these muscles in good shape. If these muscles are not working to their full capacity, it will affect the function of the other muscles, and your activity level and performance will naturally decrease.

It is important to warm up these muscles before performing any type of trunk flexion movement. Executing this stretch in a dynamic (ballistic) manner will definitely be helpful. This also decreases the possibility of injury or discomfort in these muscle groups during activity.

DYNAMIC TRUNK ROTATOR STRETCH



Execution

1. Stand upright with your feet shoulder-width apart; you can also do this stretch from a seated position.
2. Bend your elbows and place your hands close to your chest. Keep your arms in this position during this stretch.
3. With the help of your arms, rotate your torso toward each side, back and forth in a dynamic manner.
4. Keep the trunk upright, and allow the dynamic movement to take place in the trunk.

Muscles Stretched

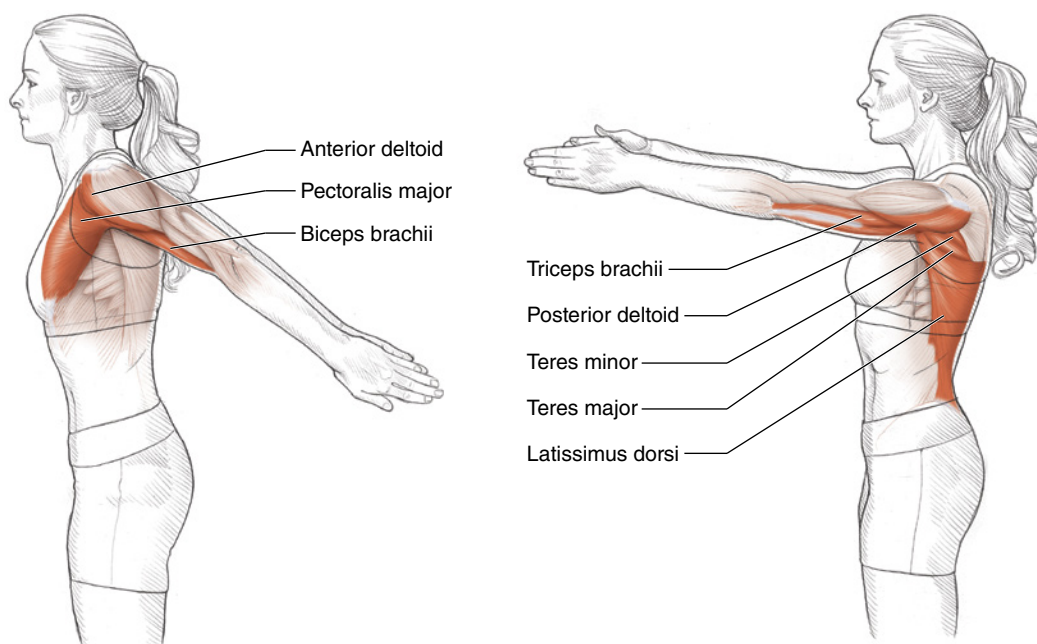
Most-stretched muscles: Multifidus, rotatores, external oblique, internal oblique

Stretch Notes

The trunk is considered the core area of the body. Trunk rotation is a common movement in many sports as well as in common household activities. You bend your trunk regularly in daily activities, perhaps hundreds of times a day. No wonder you might encounter muscular problems in this area. In addition, numerous sporting activities such as golf, tennis, and throwing sports require twisting of the trunk.

Twisting the trunk involves the trunk extensors, flexors, and lateral flexors. Improved range of motion of all lower-trunk muscles can increase the range of motion in trunk rotation and improve performance in activities that involve these actions. Warming up these muscles before any type of trunk rotation movement will definitely be helpful. Executing this stretch in a dynamic (ballistic) manner will also imitate the specific movement patterns experienced in these activities. This decreases the possibility of injury or discomfort in these muscle groups during the activity.

DYNAMIC SHOULDER FLEXION AND EXTENSION STRETCH



Execution

1. Stand upright with your feet shoulder-width apart and your arms hanging down next to your hips.
2. Swing your arms forward and backward in a dynamic manner as far as you can through the full range of motion.
3. Keep the trunk upright, and allow the dynamic movement to take place in the shoulder joint.

Muscles Stretched

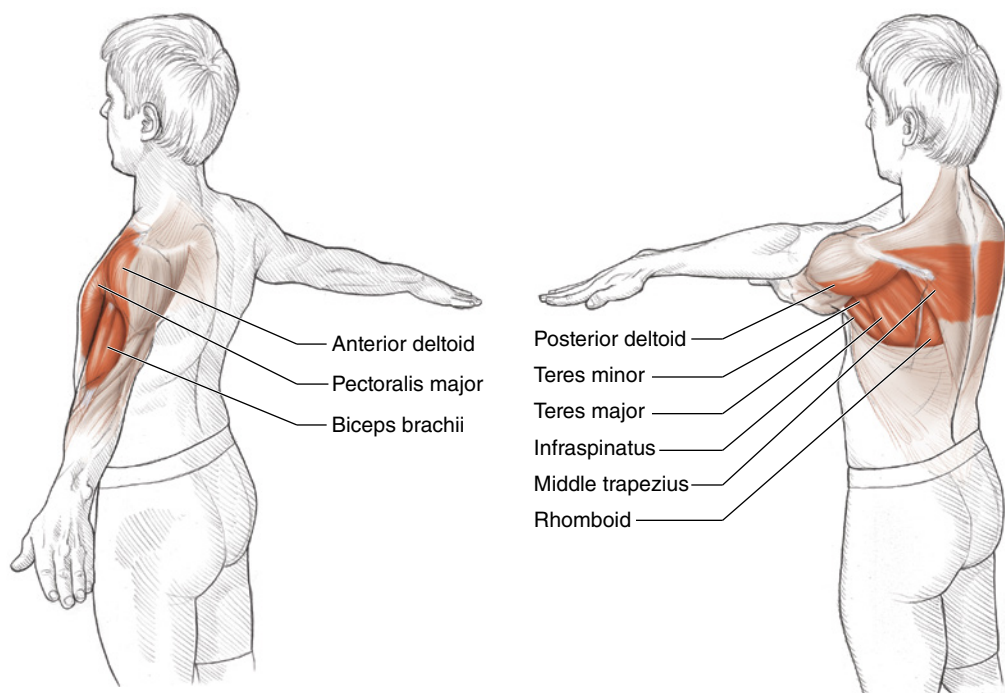
Most-stretched muscles in forward arm movement: Posterior deltoid, latissimus dorsi, teres major, teres minor, triceps brachii

Most-stretched muscles in backward arm movement: Biceps brachii, coracobrachialis, anterior deltoid, pectoralis major

Stretch Notes

You use these muscles extensively whenever you participate competitively or recreationally in any activities requiring under- or overhand throwing. Those who participate in these activities seasonally rather than all year long tend to encounter tightness or soreness in the shoulder. This is a great preexercise stretch that should be performed whenever you have tightness or aches in these muscles. This warm-up stretch is also a good way to loosen up your muscles to enhance the swinging patterns found in many sporting activities involving shoulder flexion and extension. This stretch imitates the dynamic movement patterns experienced during actual training sessions when throwing objects. Regularly stretch these muscles before and after these activities to prevent even further soreness and tightness. Continue to stretch these muscle groups as long as you participate in these activities. Warming up by using this dynamic stretching movement allows the body to get ready for your workout. This decreases the possibility of injury or discomfort in these muscle groups.

DYNAMIC SHOULDER GIRDLE ABDUCTION AND ADDUCTION STRETCH



Execution

1. Stand upright with your feet shoulder-width apart.
2. Extend your arms out to the side, slightly lower than shoulder height.
3. Swing your arms laterally back and forth in front of the chest, bringing them in medially toward your body as far as you can so they cross over each other.
4. Keep the trunk upright, and allow the dynamic movement to take place in the shoulder joint.

Muscles Stretched

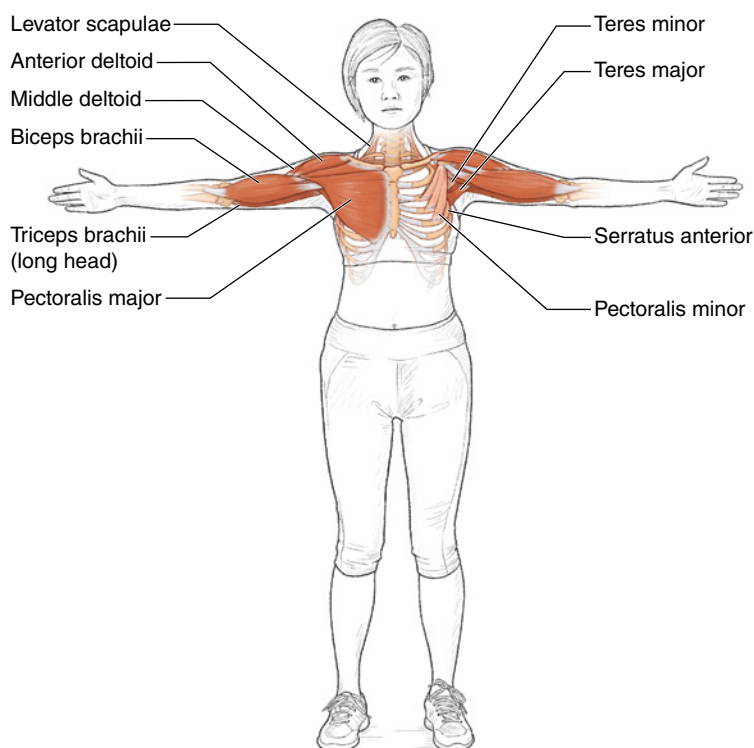
Most-stretched muscles during outward swing: Pectoralis major, pectoralis minor, anterior deltoid, coracobrachialis, biceps brachii

Most-stretched muscles during inward swing: Middle trapezius, rhomboids, posterior deltoid, teres major, teres minor, infraspinatus, supraspinatus

Stretch Notes

This stretch is a great preexercise movement for people who recreationally or competitively play any type of game that requires a racket, such as tennis, badminton, squash, and racquetball. This stretch relieves aches and tightness between the shoulder blades as well as in the chest. Also, this is a good way to loosen up your swing patterns and bring smoothness to your performance. This stretch warms these muscles in order to get rid of preexercise soreness or tightness, and it imitates the dynamic movement patterns experienced during training sessions. Warming up by using this dynamic stretching movement allows the body to get ready for your workout. It is always beneficial to perform a series of light stretches before starting any type of exercise, sport, or strenuous activity. These light stretches decrease the possibility of injury or discomfort in these muscle groups.

DYNAMIC SHOULDER CIRCUMDUCTOR STRETCH



Execution

1. Stand with the feet shoulder-width apart.
2. Hold the arms straight out to the sides and nearly parallel to the floor.
3. Rotate the arms so that the thumbs point skyward.
4. Circle the arms forward using small, controlled motions.
5. Gradually make the circle larger until a stretch is felt in the shoulder and armpit area.
6. Repeat the exercise with the arms circling in the reverse direction (circle backward).

Muscles Stretched

Most-stretched muscles: Biceps brachii, coracobrachialis, deltoid (anterior, middle, and posterior), infraspinatus, latissimus dorsi, pectoralis major, supraspinatus, teres major, teres minor, trapezius, triceps brachii (long head)

Less-stretched muscles: Pectoralis minor, levator scapulae, rhomboid, serratus anterior, splenius capitis, subscapularis, subclavius

Stretch Notes

People who spend much of the day in front of the computer or who participate competitively or recreationally in activities requiring under- or overhand throwing often experience shoulder pain. This exercise alleviates shoulder pain. Arm circles help mitigate the pain by moving the shoulder through a full range of motion. In addition to stretching the muscles, it can help to strengthen them to a mild extent.

It is important to maintain proper posture while performing this stretch. Stand up straight with shoulders back and elevated (no rounded shoulders). Improper posture can place excessive force on the entire rotator cuff and joint capsule. Depressed and rounded shoulders can also increase the chance of shoulder impingement injuries. Never lift the arms above shoulder level or lift them high enough to inflict pain. Also, make sure the size of the arm circles doesn't cause pain. If this exercise is painful or if the arms become heavy during the exercise, shorten the lever length by placing each hand on its corresponding shoulder and circling with the elbow.

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